MACHINE LEARNING

1. Movie Recommendation systems are an example of:

a) 2 Only

2. Sentiment Analysis is an example of:

d) 1, 2 and 4

3. Can decision trees be used for performing clustering?

a) True

4. Which of the following is the most appropriate strategy for data cleaning before performing clustering analysis, given less than desirable number of data points:

a) 1 only

5. What is the minimum no. of variables/ features required to perform clustering?

b) 1

6. For two runs of K-Mean clustering is it expected to get same clustering results?

b) No

7. Is it possible that Assignment of observations to clusters does not change between successive iterations in K-Means?

a) Yes

8. Which of the following can act as possible termination conditions in K-Means?

d) All of the above

9. Which of the following algorithms is most sensitive to outliers?

a) K-means clustering algorithm

10. How can Clustering (Unsupervised Learning) be used to improve the accuracy of Linear Regression model (Supervised Learning):

d) All of the above

11. What could be the possible reason(s) for producing two different dendrograms using agglomerative clustering algorithms for the same dataset?

d) All of the above

12. Is K sensitive to outliers?

Yes, K- mean is sensitive to outliers, as it uses mean of clusters data points to find the cluster centre.

Example:

The mean of 2,2,2,3,3,3,4,4,4 is 3. If we add a single 23 to that, the mean becomes 5, which is larger than *any* of the other values.

Since in k-means, you'll be taking the mean a lot, you wind up with a lot of outlier-sensitive calculations.

That's why we have the k-medians algorithm. It just uses the median rather than the mean and is less sensitive to outliers.

13. Why is K means better?

- It is simple, highly flexible, and efficient.
- The flexibility of k-means allows for easy adjustment if there are problems.
- The efficiency of k-means implies that the algorithm is good at segmenting a dataset.
- An instance can change cluster (move to another cluster) when the centroids are recomputed
- Easy to interpret the clustering results.

14. Is K means a deterministic algorithm?

K-Means is a non-deterministic algorithm. This means that a compiler cannot solve the problem in polynomial time and doesn't clearly know the next step. This is because some problems have a great degree of randomness to them.

These algorithms usually have 2 steps — 1) Guessing step 2) Assignment step. On similar lines is the K-means algorithm. The K-Means algorithm divides the data space into K clusters such that the total variance of all data points with respect to the cluster mean is minimized.